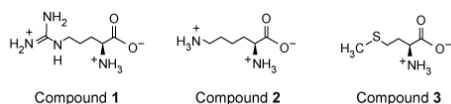


Polyamines are important biologically active molecules produced *in vivo* by a pathway that utilizes the amino acids shown below. Which of the following amino acid abbreviations does NOT correspond to one of the given compounds?



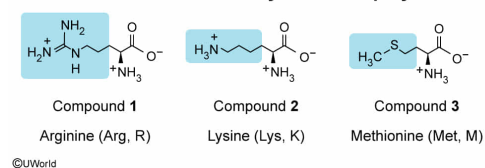
- ☐ A. Arg [4%]
- ✓ ☒ B. Asn [88%]
- ☐ C. Met [3%]
- ☐ D. Lys [4%]

Correct

87% answered correctly

Explanation:

Amino acids used in the synthesis of polyamines



Amino acids are important biological molecules that make up proteins. The **structure** of an amino acid consists of an **amino group**, a **carboxyl group**, and a **side chain**, all attached to a central carbon atom (ie, the **α-carbon**). Based on the nature of their side-chain functional group, amino acids are classified as either **hydrophobic** (nonpolar) or **hydrophilic** (polar). The polar amino acids can be further classified as **acidic** (ie, negatively charged at physiological pH), **basic** (positively charged), or **uncharged**. In addition to their chemical names, amino acids have **one-letter and three-letter abbreviations**.

Asparagine (Asn, N) is a polar amino acid containing an **amide** side chain. The structures of Compounds **1–3** given in the question do not contain an amide functional group. Therefore, Asn is *not* one of the given compounds and is not used in the synthesis of polyamines.

(Choice A) Arginine (Arg, R) is a basic amino acid containing a **guanidinium group**, which is present in Compound **1**.

(Choice C) **Methionine** (Met, M) is a nonpolar amino acid containing a **thioether** side chain, which is shown in Compound **3**.

(Choice D) **Lysine** (Lys, K) is a basic amino acid that contains a four-carbon chain with a **primary amine** at the end of its side chain. This description corresponds to Compound **2**.

Educational objective:

Amino acids are biological molecules that make up proteins; they contain an amino group, a carboxyl group, and a side chain all bonded to a central carbon atom (ie, the α-carbon). Amino acids are classified in several ways (eg, hydrophilic or hydrophobic, acidic or basic, charged or uncharged) based on the side-chain functional group. In addition to their chemical names, amino acids have one-letter and three-letter abbreviations.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 403140

System:
Functional Groups and Their Reactions